

19 Poisson Distributions

- Binomial and Negative Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location

Example 19.1. Recall the matching problem with a general n : there are n objects that are shuffled and placed uniformly at random in n spots with one object per spot. Note: n is a fixed number, but we are considering different values of it.

We have seen that $E(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$) the approximate distribution of X is the Poisson(1) distribution.

1. Does X have a Binomial distribution? Consider: What is a trial? What is success? Is the number of trials fixed? Is the probability of success the same on each trial? Are the trials independent?
2. Starting with $x = 0$, describe the distributon of X in terms of relative likelihoods: “1 is [blank] times as likely as 0”, “2 is [blank] times as likely as 1”, etc
3. Specify the pmf of X .

Table 19.1: Poisson(1) distribution, the approximate distribution of X , the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.3679
1	0.3679
2	0.1839
3	0.0613
4	0.0153
5	0.0031
6	0.0005
7	0.0001

4. Use the pmf to compute $E(X)$.

5. Use the pmf to compute $\text{Var}(X)$.

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, \dots$ and the pmf follows the

Poisson(μ) pattern: x is $\frac{\mu}{x}$ as likely as $x - 1$ (for $x = 1, 2, 3, \dots$)

- As a function of $x = 0, 1, 2, \dots$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
 - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
 - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.
- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The shape of a Poisson pmf as a function of x is given by $\mu^x/x!$. The constant $e^{-\mu}$ simply renormalizes the heights of the pmf so that the probabilities sum to 1.
- If X has a $\text{Poisson}(\mu)$ distribution then

$$\begin{aligned} E(X) &= \mu \\ \text{Var}(X) &= \mu \end{aligned}$$

Example 19.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a $\text{Poisson}(2.4)$ distribution.

1. In what ways is this like the Binomial situation?
2. In what ways is this NOT like the Binomial situation?
3. In what sense are home runs “relatively rare”?
4. Interpret the parameter 2.4 in context.
5. Compute $P(X = 2.4)$.
6. Identify the most likely value of X .

7. Specify the pmf of X .

8. Compute $P(X = 0)$ and interpret the value in context.

9. Construct a table and spinner corresponding to the distribution of X .

Example 19.3. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a $\text{Poisson}(0.53)$ distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a “month”).

1. What is $E(X)$? You should be able to compute it without knowing the distribution of X .

2. Compute $P(X = 0)$.

3. Compute $P(X = 1)$.

4. How could you use simulation to approximate the distribution of X ?

5. Use simulation to approximate the distribution of X . Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $P(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a $\text{Poisson}(\mu_X)$ distribution, and Y has a $\text{Poisson}(\mu_Y)$ distribution, then $X + Y$ has a $\text{Poisson}(\mu_X + \mu_Y)$ distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

Example 19.4. Recall the birthday problem: in a group of n people what is the probability that at least two have the same birthday? (Ignore multiple births and February 29 and assume that the other 365 days are all equally likely.) We will investigate this problem using Poisson approximation. Imagine that we have a trial for each possible *pair* of people in the group, and let “success” indicate that the pair shares a birthday. Consider both a general n and $n = 30$.

1. How many trials are there?

2. Do the trials have the same probability of success? If so, what is it?

3. Are any *two* trials independent? To answer this questions, suppose that three people in the group are Bill, Frank, and Julia and consider any *two* of the trials that involve these three people.

4. Are any *three* trials independent? Consider the three trials that involve Bill, Frank, and Julia.

5. Let X be the number of *pairs* that share a birthday. Does X have a Binomial distribution?

6. If X has an approximate Poisson distribution, what would the parameter have to be? Compare this Poisson distribution with a simulation of X ; does it seem like a reasonable approximation?

7. Approximate the probability that at least two people share the same birthday. Compare to the theoretical value.

Poisson paradigm. Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = P(A_i)$. Let $N = I_{A_1} + I_{A_2} + \dots + I_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose

- n is “large”,
- $p_1, \dots, p_n$ are “comparably small”, and
- the events $A_1, \dots, A_n$ are “not too dependent”,

Then N has an approximate Poisson distribution with parameter $E(N) = \sum_{i=1}^n p_i$.

Example 19.5. Use Poisson approximation to approximate that probability that at least *three* people in a group of n people share a birthday. How large does n need to be for the probability to be greater than 0.5?